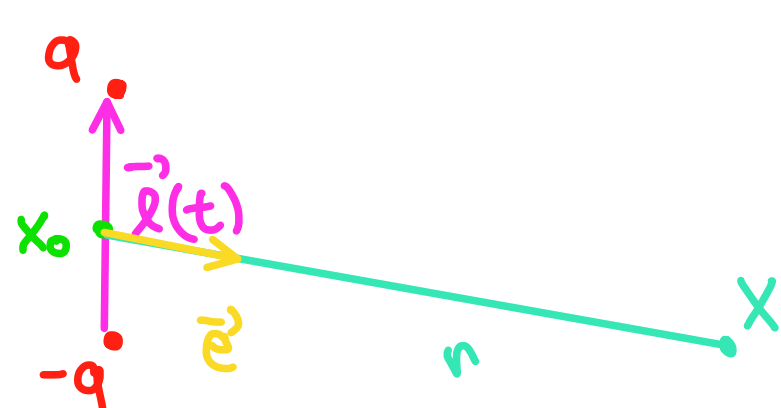


Dipólové záření



$$\vec{d}(t) = q\vec{l}(t)$$

$$\rho = -\vec{d} \cdot \vec{\nabla} \delta_{x_0}$$

$$\vec{j} = \dot{\vec{d}} \delta_{x_0}$$

$$x_0 = 0$$

$$\vec{A} = \frac{\mu}{4\pi} \int \frac{\vec{j}(t - \frac{r(x|x')}{c})}{r(x|x')} dX' = \frac{\mu}{4\pi} \frac{\dot{\vec{d}}(t - \frac{r}{c})}{r}$$

$$\phi = \frac{1}{4\pi\epsilon} \int \frac{\rho(t - \frac{r(x|x')}{c}, x')}{r(x|x')} dX' \stackrel{\text{přechodem}}{=} \vec{\nabla}'$$

$$= \frac{1}{4\pi\epsilon} \int \left[\vec{\nabla}' \cdot \frac{\vec{d}(t - \frac{r(x|x')}{c})}{r(x|x')} \right] \delta_0 dX' \stackrel{\text{je vektor (přeslechl)}}{\vec{\nabla}' r = -\vec{e}}$$

$$= \frac{1}{4\pi\epsilon} \left[\frac{\dot{\vec{d}}(t - \frac{r}{c}) \cdot (-\frac{1}{c})(-\vec{e})}{r} - \frac{\vec{d}(t - \frac{r}{c}) \cdot \vec{e}}{r^2} \right]$$

$$= \frac{1}{4\pi\epsilon} \left[\frac{\dot{\vec{d}}(t - \frac{r}{c}) \cdot \vec{e}}{rc} - \frac{\vec{d}(t - \frac{r}{c}) \cdot \vec{e}}{r^2} \right]$$

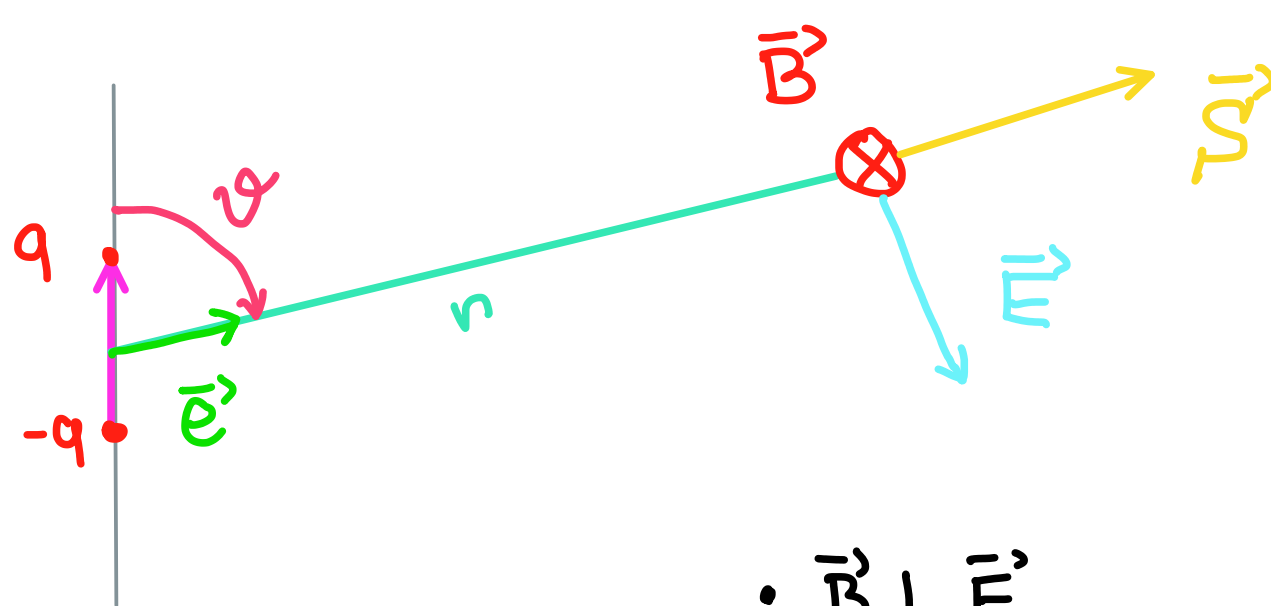
Chceme $\vec{E}$ a $\vec{B}$, ale máme jen členy $\propto \frac{1}{r}$,
jiní v dlece vynecháme $\vec{\nabla} \vec{e} \sim \frac{1}{r^2}$

$$\vec{B} = \vec{\nabla} \times \vec{A} = -\frac{\mu}{4\pi} \frac{\vec{e} \times \ddot{\vec{d}}(t - \frac{r}{c})}{cr} + O(\frac{1}{r^2}) \quad \frac{1}{4\pi\epsilon c^2}$$

$$\begin{aligned} \vec{E} &= -\vec{\nabla} \phi - \frac{\partial \vec{A}}{\partial t} = -\frac{1}{4\pi\epsilon} \left(-\frac{\vec{e} \ddot{\vec{d}}(t - \frac{r}{c}) \cdot \vec{e}}{c^2 r} \right) - \frac{\mu}{4\pi} \frac{\ddot{\vec{d}}(t - \frac{r}{c})}{r} \\ &= +\frac{1}{4\pi\epsilon c^2} \frac{\vec{e} \times [\vec{e} \times \ddot{\vec{d}}(t - \frac{r}{c})]}{r} + O(\frac{1}{r^2}) \end{aligned}$$

$$\vec{B} = -\frac{1}{4\pi\epsilon c^2} \frac{\vec{e} \times \ddot{\vec{d}}(t - \frac{r}{c})}{cr} + O(\frac{1}{r^2})$$

$$\vec{E} = \frac{1}{4\pi\epsilon c^2} \frac{\vec{e} \times [\vec{e} \times \ddot{\vec{d}}(t - \frac{r}{c})]}{r} + O(\frac{1}{r^2})$$



$$\vec{B} \perp \vec{E}$$

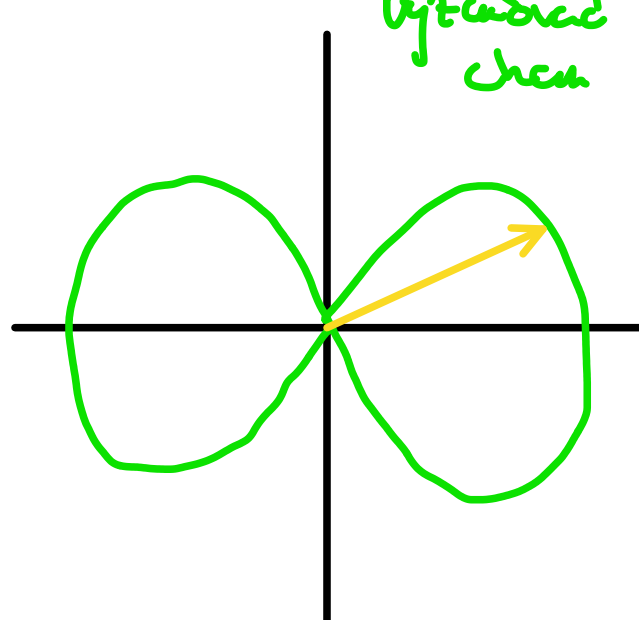
$$E = cB$$

$$\vec{S} = \vec{E} \times \vec{H} = \epsilon c^2 \vec{E} \times \vec{B} = \epsilon_0 c^3 B^2 \vec{e}$$

$$= \frac{1}{(4\pi)^2 \epsilon c^3} \frac{|\vec{e} \times \ddot{\vec{d}}_{\text{vet}}|^2}{r^2} \vec{e}$$

výživoval chom

$$\vec{S} = \frac{1}{(4\pi)^2 \epsilon c^3} \frac{\ddot{d}_{\text{vet}}^2 \sin^2 \vartheta}{r^2} \vec{e}$$



→ rychlost vyžarování

$$Q = \int_{\text{velká}} \vec{S} \cdot d\vec{S} = \frac{\ddot{d}_{\text{vet}}^2}{(4\pi)^2 \epsilon c^3} \int \frac{\sin^2 \vartheta}{r^2} r^2 \sin \vartheta d\vartheta d\varphi \quad 2\pi \frac{4}{3}$$

$$Q = \frac{2}{3} \frac{\ddot{d}_{\text{vet}}^2}{4\pi \epsilon c^3}$$